

Summarizing Data II

Math 11: Lecture 2

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Dataset: an example

Academic salaries data:

Variables

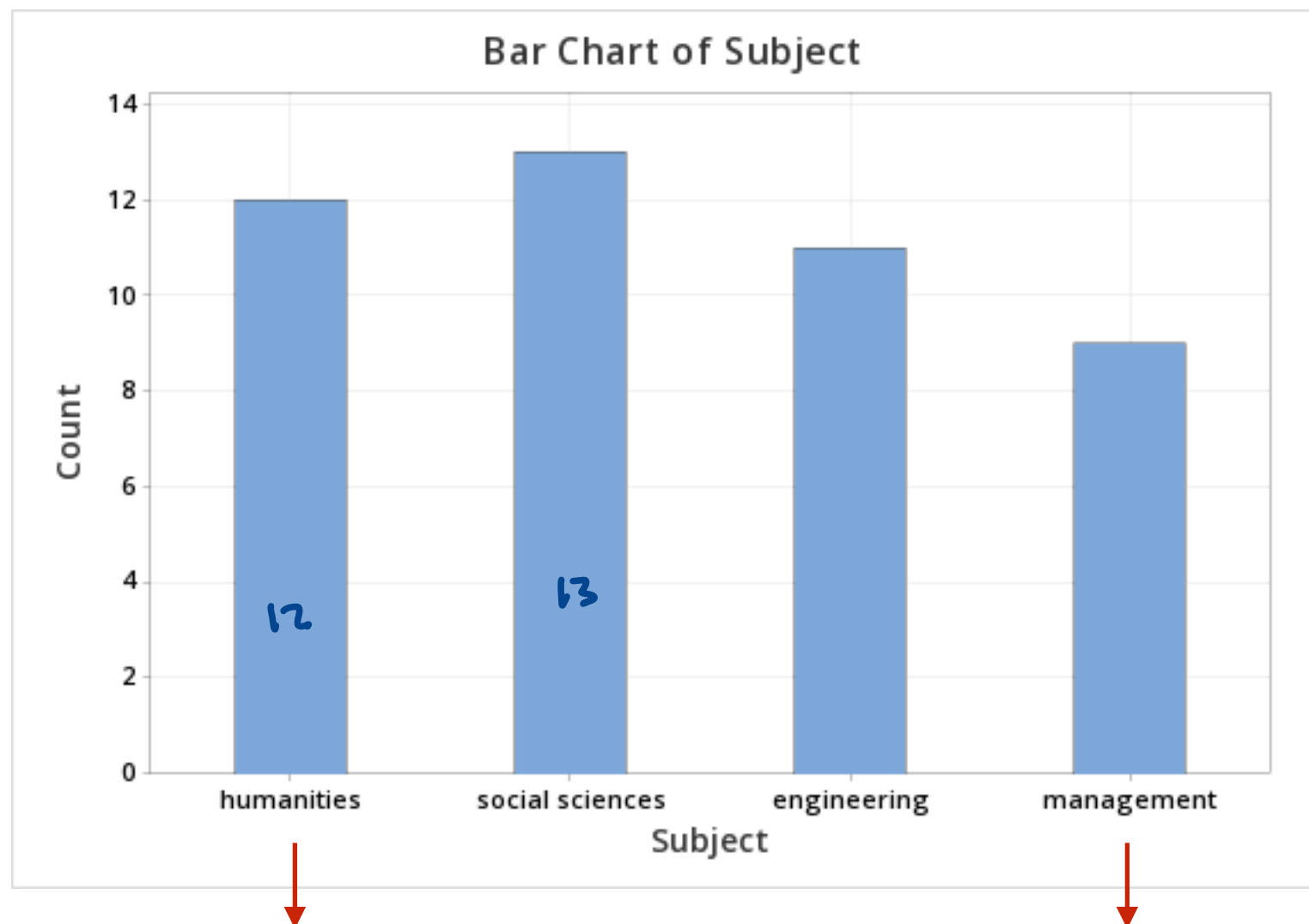
Subject	Degree	Salary
management	bachelor's	2.5
humanities	Ph.D.	2.6
social sciences	bachelor's	2.6
social sciences	master's	2.6
management	bachelor's	2.6
humanities	Ph.D.	2.7
humanities	Ph.D.	2.7
social sciences	master's	2.7
engineering	bachelor's	2.7
engineering	master's	2.7

Observations

Source: <https://support.minitab.com/en-us/datasets/anova-data-sets/academic-salaries/>

Visualizing Data

Categorical/Qualitative: **Bar Chart**.

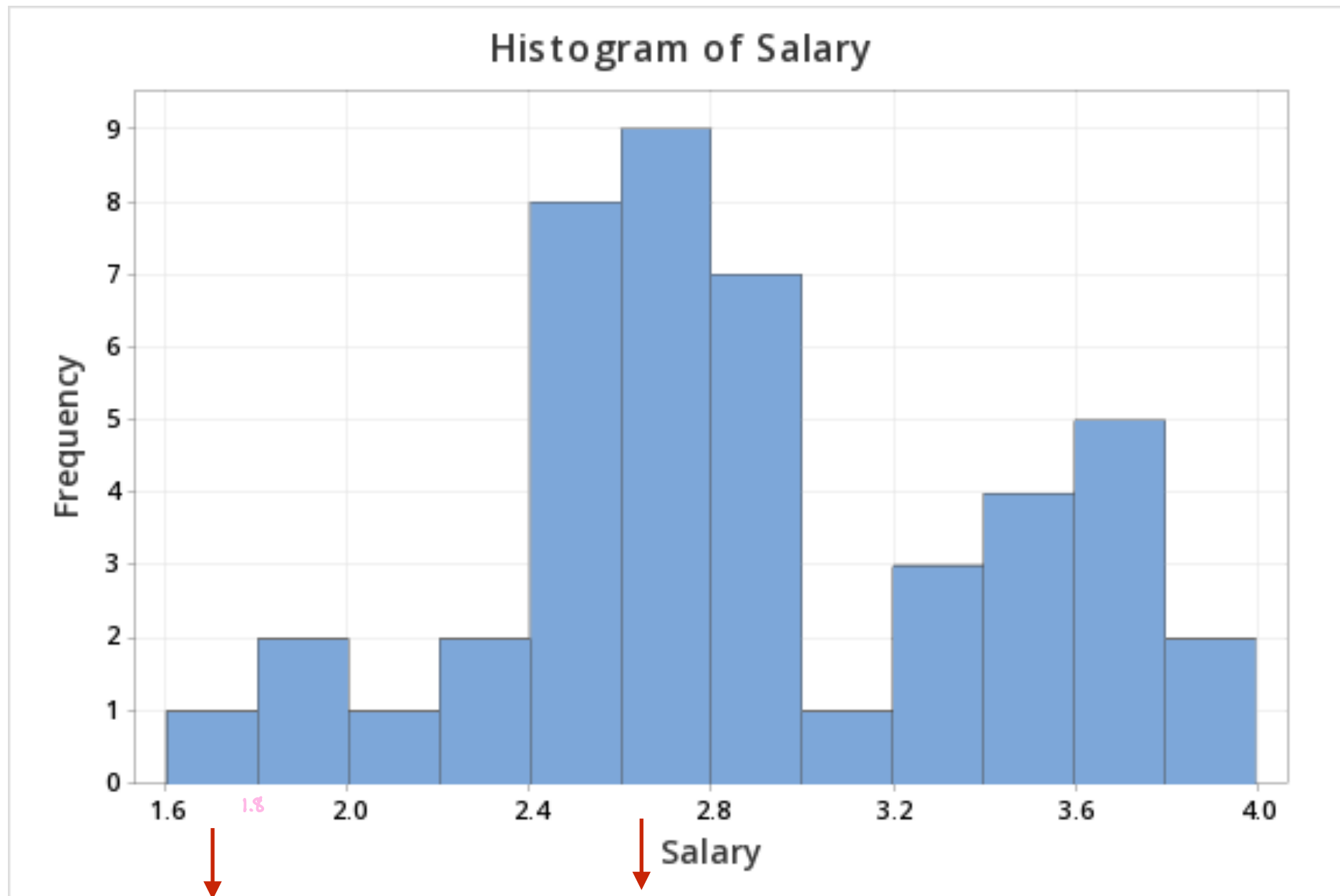


12 professors teach humanities' courses

9 professors teach management' courses

Visualizing Data

Numeric/Quantitative: **Histogram.**

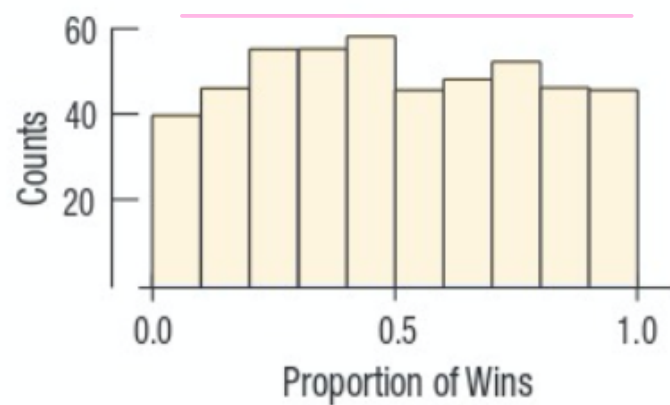


Only one professor got a salary between 1.6 and 1.8.

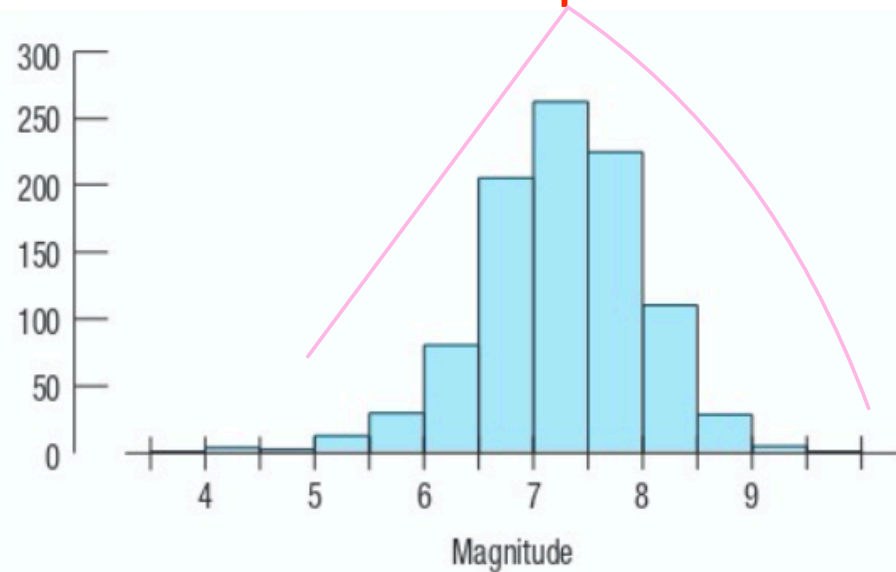
9 professors got a salary between 2.6 and 2.8.

Distribution description

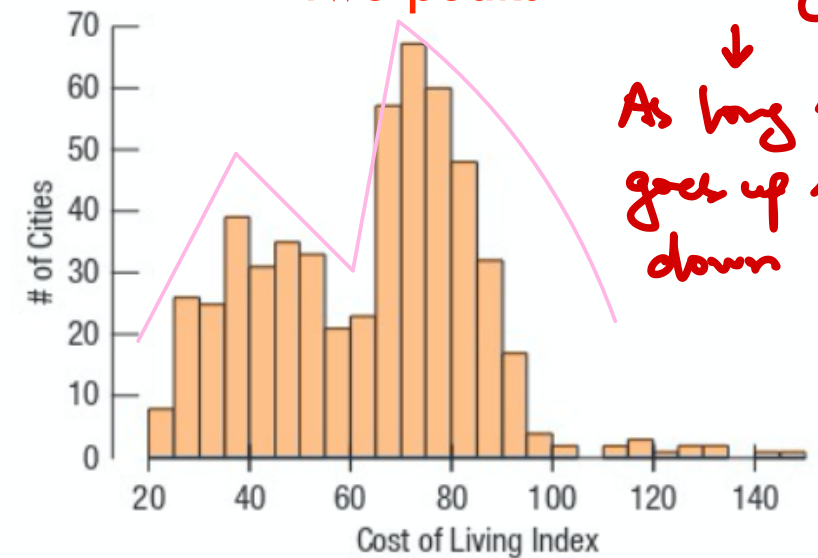
Uniform
No peaks



Unimodal
One peak

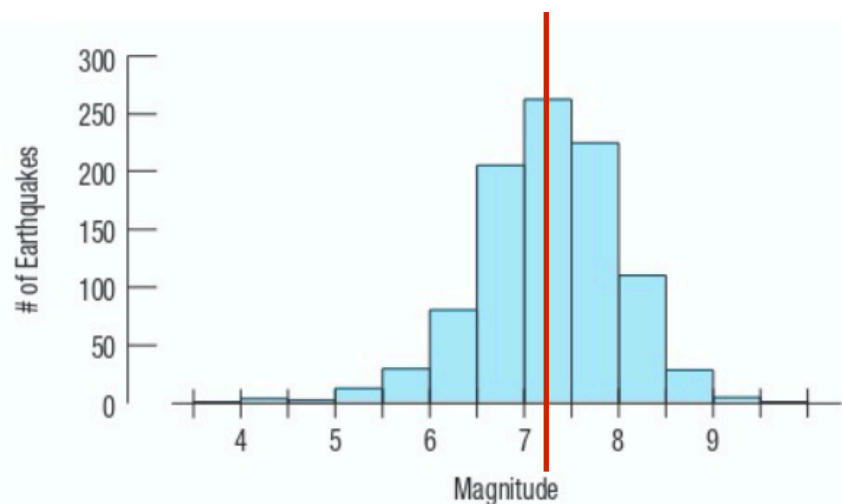


Bimodal
Two peaks



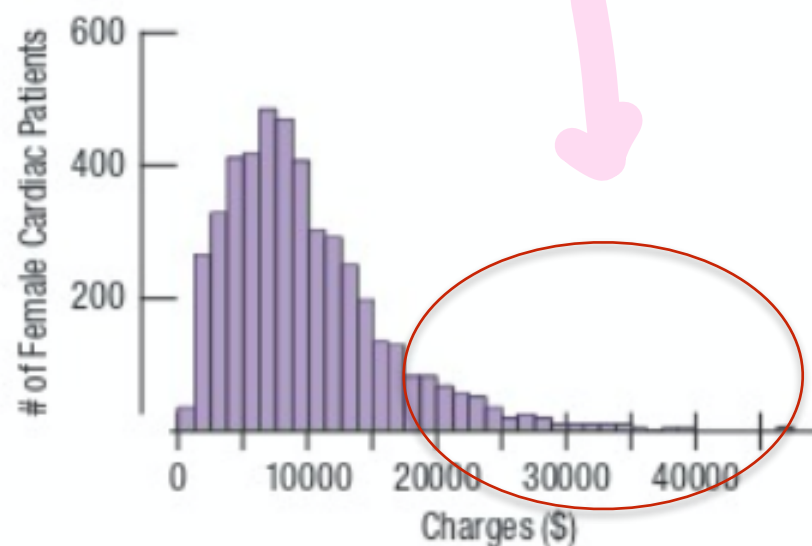
Peaks don't
have to be the
same height!
↓
As long as data
goes up & then
down

Symmetric



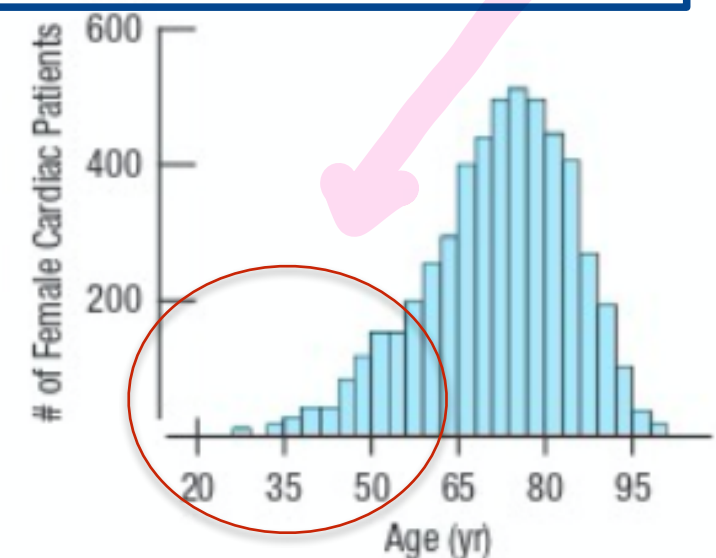
ASymmetric UNIMODAL

Skewed to the right

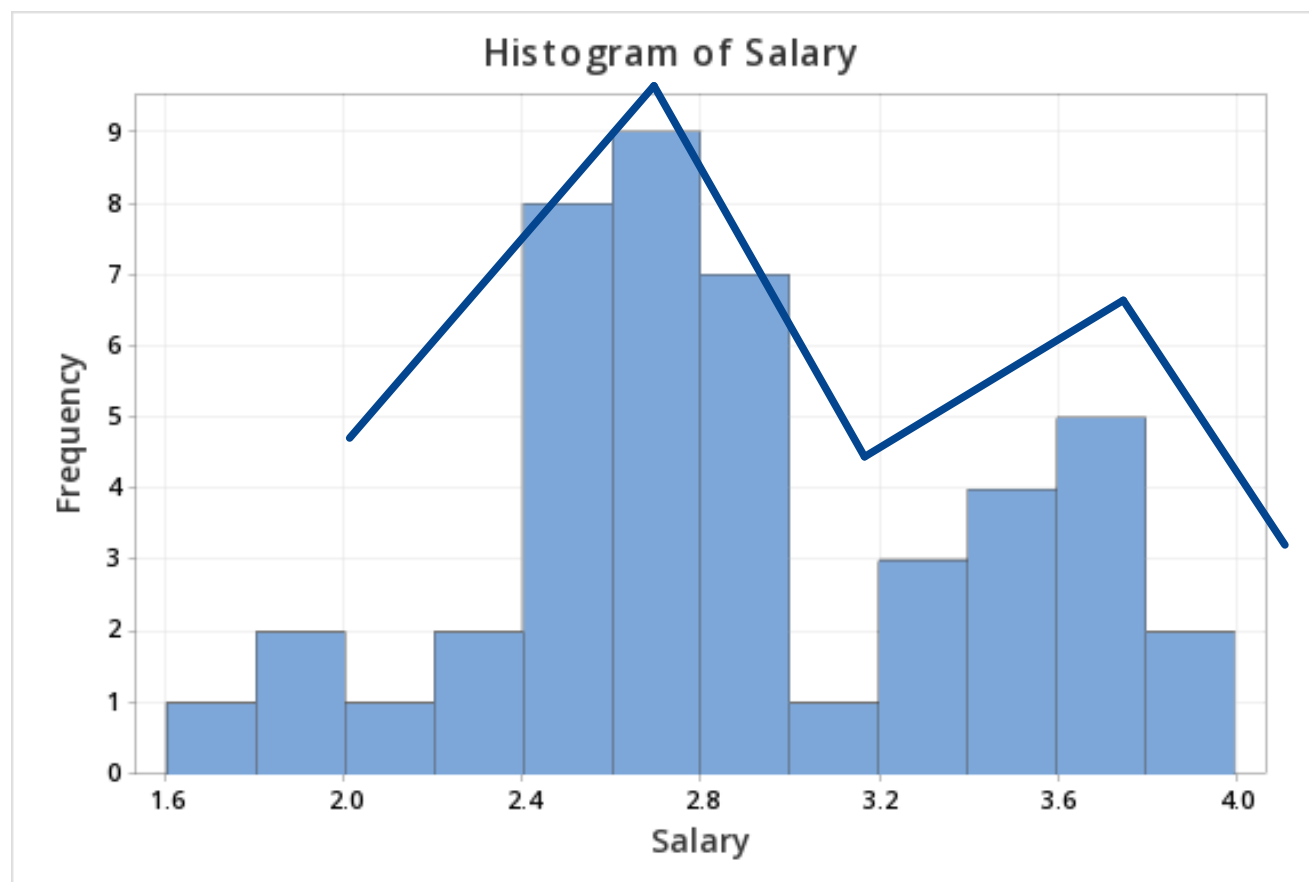


MULTIMODAL: More than 3 peaks

Skewed to the left



Distribution description



Bimodal
Asymmetric

2 peaks → Bimodal
/ ONE mound 2.7
ONE mound 3.6

RANGE: 1.6 & 4.0
/ x-axis
y-axis = counter

Summarizing Data

Categorical/Qualitative: **Contingency table.**

Subject	Count
humanities	12
social sciences	13
engineering	11
management	9
N=	45

Tells how many elements are there

Numeric/Quantitative: **Measures of center and of spread.** *Next slides*

Measures of center for numeric variable

How much money a professor tends to make?

- **Mean:** average of data.

{1, 3, -5, 4, -2}

$$\bar{X} = \frac{1+3-5+4-2}{5} = .2$$

↑
mean annotation

Median
Middle

Sum all the observations and divide by the number of the observations

- **Median:** value in the middle of the observations.

{1, 3, -5, 4, -2} ~~-5~~ ~~-2~~ 1 ~~3~~ ~~4~~

↑
Median

Arrange the observations in increasing order.

If you have an odd number of observations pick the value in the middle.

{1, 3, -5, 4, -2, 8} ~~-5~~ ~~-2~~ 1 3 ~~4~~ ~~8~~

TAKE THE AVERAGE

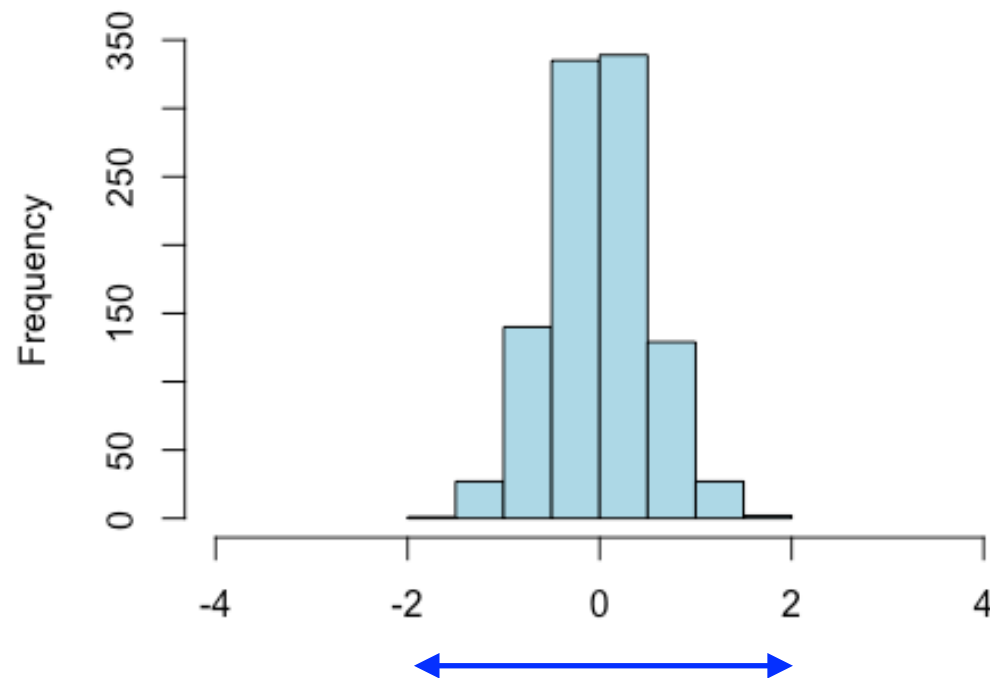
↓
1+3
—
2 = 2

Median
↓
2

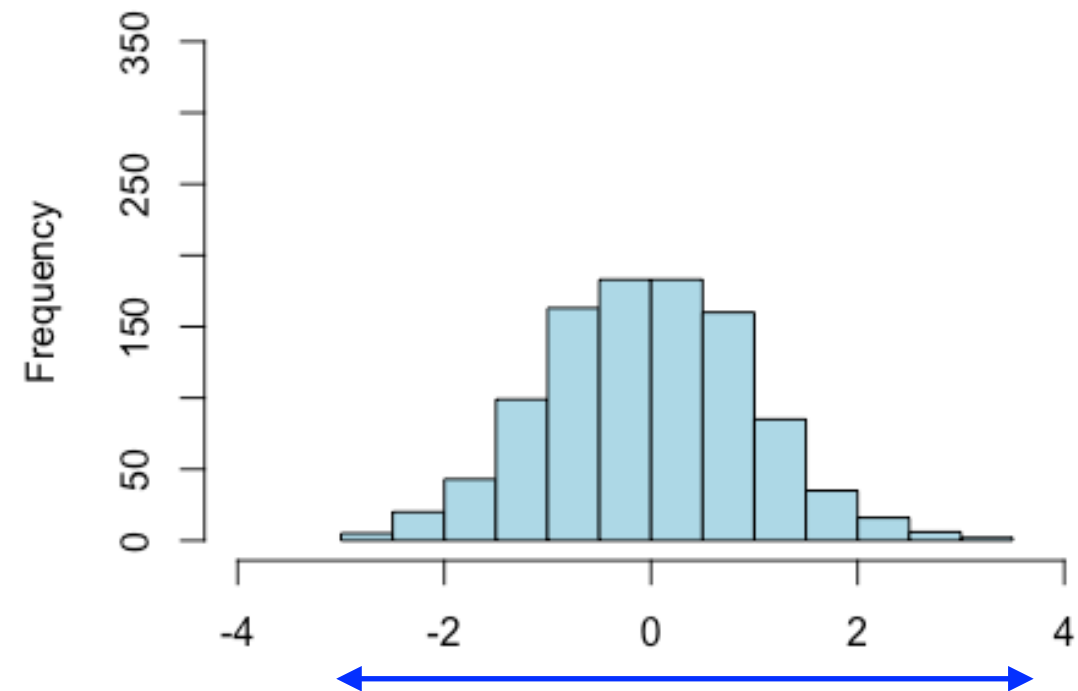
If you have an even number of observations average the two values in the middle.

when data's symmetric, mean & median
tend to be around same number

Same mean median but...



Mean: -0.0004476649
Median: -0.0007694833



Mean: -0.03220894
Median: -0.0482122

The two data have the same measures of center, however their histograms look different. In the first, the data are more concentrated around the mean, median. This behavior can be summarized by the **measures of spread**.

Measures of spread for numeric variable

How close is the salary of a college professor to its mean/median?

• **Variance:** $\{x_1, \dots, x_n\}$

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

↑
variance symbol

Measure how far each observation is from the mean.

Measuring how far each data is from the mean ($\bar{x}$)

Standard deviation:

$$s = \sqrt{s^2}$$

Standard deviation is the square root of the variance.

$$\{1, 3, -5, 4, -2\} \text{ — } \bar{x} = .2$$

$$s^2 = \frac{(1-.2)^2 + (3-.2)^2 + (-5-.2)^2 + (4-.2)^2 + (-2-.2)^2}{4} = 13.7$$

$$s = \sqrt{13.7} = 3.701$$

↑
Larger the number,
the bigger the spread

Measures of spread for numeric variable

Q_1 = Marker of 25% (25% prior, 75% after)

Q_3 = Marker of 75% (75% prior, 25% after)

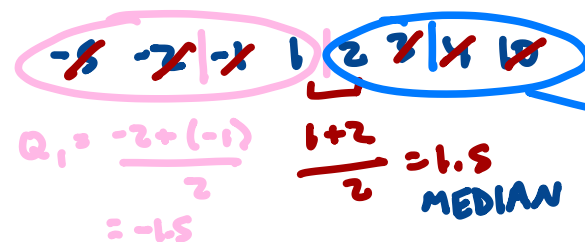


Q_1, Q_3 are the first and the third quartile, i.e. the values that mark the 25% and the 75% of the ordered observations.

- **Interquartile range (IQR):** $Q_3 - Q_1$

$\{1, 3, -5, 4, -2, 2, -1, 10\}$

Put the observations in order.



$$Q_3 = \frac{3+4}{2} = 3.5$$

Find the median and divide the observations in half.

The first quartile is the median of the first half (to the left of the median).

$$IQR = Q_3 - Q_1 = 3.5 - (-1.5) = 5$$

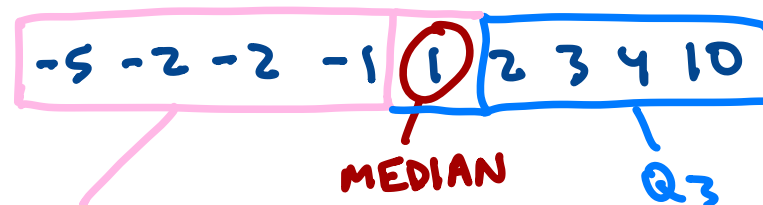
The third quartile is the median of the second half (to the right of the median).

IQR: another example

if median's in middle, can include or not
include to calculate IQR $\rightarrow$ just stay consistent

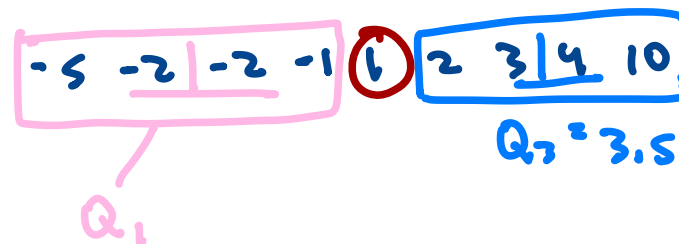
If you have more than once the same number, you have to repeat it when you order the data.

$\{1, 3, -5, 4, -2, 2, -1, 10, -2\}$



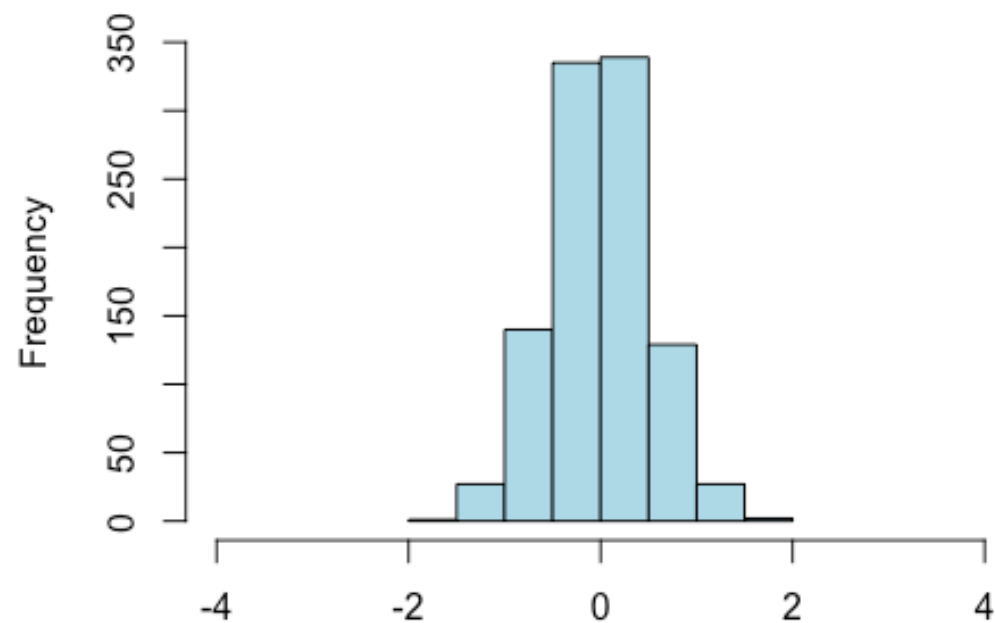
$$IQR = 3 - (-2) = 5$$

same answer!

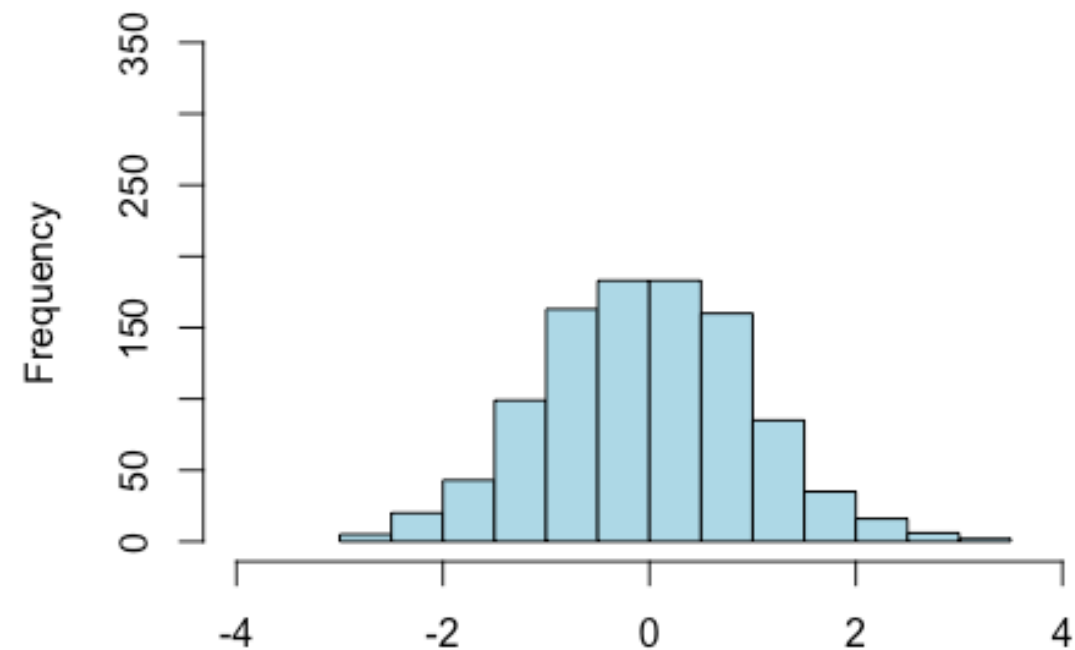


$$Q_3 = 3.5$$

Back to the example



Mean: -0.0004476649
Median: -0.0007694833
Standard deviation: 0.5086543
IQR: 0.6977639



Mean: -0.03221
Median: -0.04821
Standard deviation: 0.9975586
IQR: 1.395366

More on mean and sd

$\{1, 5, -2, 6, 3\}$

↓ +10

$\{11, 15, 8, 16, 13\}$

$$2.6 + 10 = 12.6$$

$$\bar{x} = 12.6$$

$$\bar{x} = \frac{1 + 5 - 2 + 6 + 3}{5} = 2.6$$

↑
MEAN

MEAN changes w/ data

Socrative

1. The new mean is: (Base your answer on your intuition)

Show Results

0/0 Students Answered

A 12.6

B 2.6

C -7.4

D 26

More on mean and sd

{1, 5, -2, 6, 3}

↓ +10

{11, 15, 8, 16, 13}

$$s = \sqrt{\frac{(1 - 2.6)^2 + \dots + (3 - 2.6)^2}{4}} = 3.209$$

*SD doesn't change b/c spread
doesn't change by shifting ALL data*

Socrative

2. The new sd is: (Base your answer on your intuition)

Show Results

0/0 Students Answered

 SD IS INVARIANT
TO TRANSLATION

A 13.209

☒ B 3.209

C 32.09

D None of the above

More on mean and sd

{1, 5, -2, 6, 3}

$$\bar{x} = \frac{1 + 5 - 2 + 6 + 3}{5} = 2.6$$



{2, 10, -4, 12, 6}

$\bar{x} = 5.2$ *mean is also doubled*
↓
 $\bar{x} = 2.6(2) = 5.2$

Socratic

3. The new mean is: (Base your answer on your intuition)

Show Results

0/0 Students Answered

A 4.6

B 2.6

C 5.2

D None of the above

More on mean and sd

{1, 5, -2, 6, 3}



{2, 10, -4, 12, 6}

$$s = \sqrt{\frac{(1 - 2.6)^2 + \dots + (3 - 2.6)^2}{4}} = 3.209$$

$$s = 3.209(2) = 6.418$$

SD CHANGES b/c data is becoming more spread out by $\times 2$

Socratic

4. The new sd is: (Base your answer on your intuition)

Show Results

0/0 Students Answered

A 5.209

B 3.209

☒ C 6.418

D None of the above

Data set

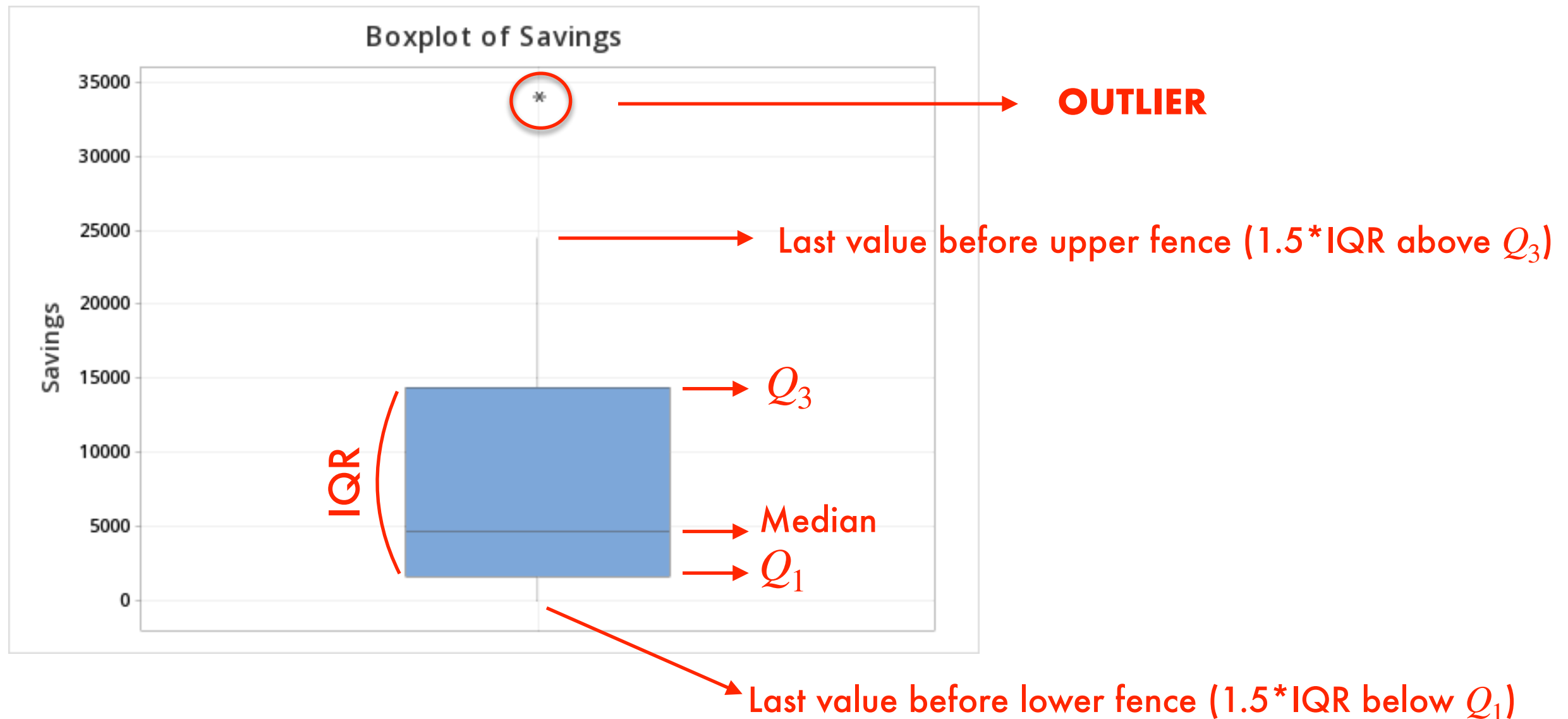
Loan applicant data:

Income	Education	Age	Residence	Employ	Savings	Debt	Credit cards
50000	16	28	2	2	5000	1200	2
72000	18	35	10	8	12000	5400	4
61000	18	36	6	5	15000	1000	2
88000	20	35	4	4	980	1100	4
91100	18	38	8	9	20000	0	1
45100	14	41	15	14	3900	22000	4
36200	14	29	6	5	100	7000	5

Source: <https://support.minitab.com/en-us/datasets/basic-statistics-data-sets/loan-applicant-data/>

Boxplot

Visualizing numeric variables: histogram or **boxplot**



Recap

- Variables can be **numeric continuous, numeric discrete, categorical ordinal or categorical nominal**.
- Displaying the distribution of numeric variables: **histogram and/or boxplot**
- Displaying the distribution of categorical variables: **bar charts**
- Summarizing numeric variables: measures of center (**mean or median**), measures of spread (**standard deviation/ variance or IQR**)
- Summarizing categorical variables: **contingency table**